# Coupling biofilm growth into an existing FEM element

Master's thesis colloquium

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16 December 2026

# Motivation and objective

- Peri-implantitis affects about one in five implant patients (prevalence 22 % and 19.5 % in two meta-analyses)
- It is driven by multi-species biofilms; a shift towards pathobionts such as *P. gingivalis* goes with disease
- A biofilm is also a mechanical body: where it grows unevenly or against a stiff substrate, it carries stresses

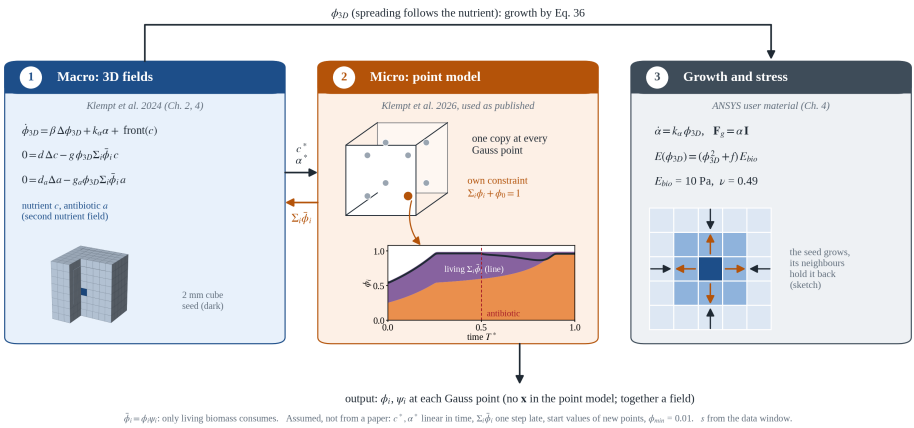
**Objective:** to compute the species composition of a biofilm, its growth and the resulting stresses consistently within an existing finite element framework.

**Point model**
species composition at a material point
no spatial resolution, no stress

**Growth field**
spatial growth and growth-induced stress
single biofilm phase, no species

**This work**
both, evaluated at every Gauss point of an existing ANSYS element

# Approach




→ The point model runs unchanged at every Gauss point; nutrient and antibiotic come from the fields, and only living biomass consumes them.

# Both models from the extended Hamilton principle

**Growth field** (Klempt et al. 2024)

$$\dot{\phi} = \beta \nabla^2 \phi + k_\alpha \alpha \ (-\text{front term})$$
$$\dot{c} = d \nabla^2 c - g \phi c, \qquad \dot{\alpha} = k_\alpha \phi$$
$$\mathbf{F} = \mathbf{F}_e \mathbf{F}_g, \quad \mathbf{F}_g = \alpha \mathbf{I}, \quad \alpha(0) = 1$$

Stress only from $\mathbf{F}_e$; growth is $\alpha - 1$.

$\phi \in [0,1]$: amount of biofilm; $c$: nutrient concentration; $\alpha$: growth variable, growth $= \alpha - 1$; $\beta$: diffusion coefficient of $\phi$ [mm$^2/T^*$]; $k_\alpha$: growth rate [$1/T^*$]; $d$, $g$: nutrient diffusivity [mm$^2/T^*$], consumption rate [$1/T^*$]; $T^*$: normalised time of Klempt et al. 2024

**Point model** (Klempt et al. 2026)

$$\Psi + \Delta + C \ \rightarrow \ \underset{\dot{\boldsymbol{\phi}}, \dot{\boldsymbol{\psi}}, \gamma}{\text{stat}}$$

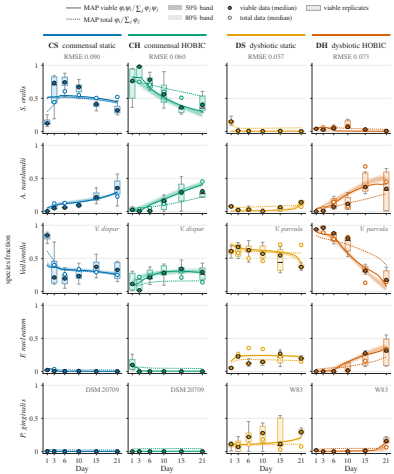
$\Psi = -\frac{1}{2} c^* \bar{\boldsymbol{\phi}}^{\mathrm{T}} \mathbf{A} \bar{\boldsymbol{\phi}}$, $\bar{\phi}_i = \phi_i \psi_i$, $C = \gamma(\sum_l \phi_l - 1)$

local form: principle of the minimum of the dissipation potential (Junker et al. 2014, Hackl 1997)

$\Rightarrow$ $\mathbf{A}$ symmetric, $2n+2$ equations per point

$\phi_i$, $\psi_i$: volume fraction and viability of species $i$; $\bar{\phi}_i$: living fraction; $\Psi$: free energy, $\Delta$: dissipation function, $C$: constraint function; $c^*$: nutrient level; $\mathbf{A}$: interaction matrix; $\gamma$: Lagrange multiplier of $\sum_{l=0}^{n} \phi_l = 1$ ($\phi_0$: void)

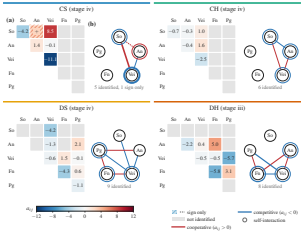
# Part 1: Bayesian calibration of the point model

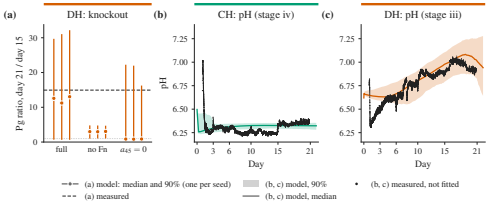



- Heine et al. (2025): five species, four conditions (commensal / dysbiotic, static / flow), days 1–21
- 15 coefficients $a_{ij}$ per condition, effective and undirected
- TMCMC in four stages, three seeds each, with convergence checks
- JAX on the GPU: about 150 times faster than the CPU

**Fit:** RMSE 0.03 to 0.18; in CS composition and live fraction cannot both be matched

# Part 1: what the data determine

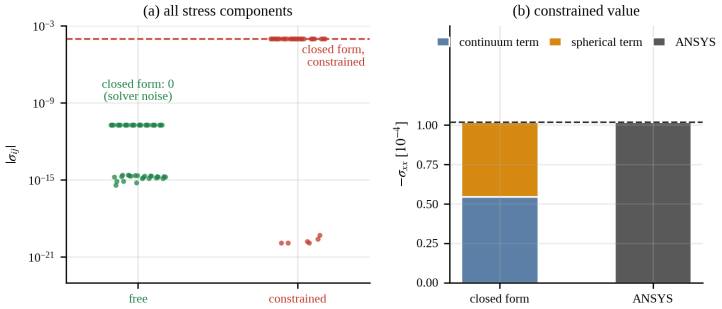



- Identified entries: 4, 6, 8, 9 of 15 (CS, CH, DS, DH)
- Commensal: only the early colonisers; *So–Vei* positive in both
- pH not used in the inference: a check, followed in DH

→ **Testable hypothesis:** without *F. nucleatum*, the late growth of *P. gingivalis* under dysbiotic flow disappears in the model.

# Part 2: implementation of the growth law in ANSYS




**Material**

ANSYS user material, $\mathbf{F}_g = \alpha\mathbf{I}$, neo-Hookean

**Stiffness (Klempt et al. 2024)**

$E(\phi) = (\phi^2 + f)E$, $E = 10$ Pa, $\nu = 0.49$; $f = 10^{-3}$ (this work)

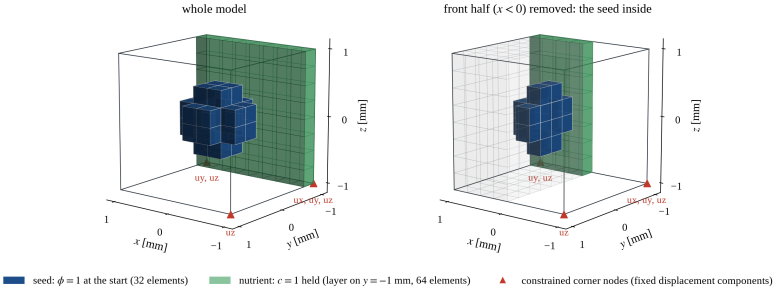
**Check (one element)**

free: zero stress; constrained: $-1.0193 \times 10^{-4}$ MPa, as predicted

$\rightarrow$ The implementation reproduces the closed-form solutions for free (stress-free) and constrained growth.

# Part 2: coupling to the IKM element

ANSYS model: cube of 2 mm, $8 \times 8 \times 8 = 512$ hexahedral elements of 0.25 mm




**Model**

2 mm cube, $8^3$ SOLID185 (B-bar), seed of 32 elements; also $16^3$, $24^3$

**Unchanged**

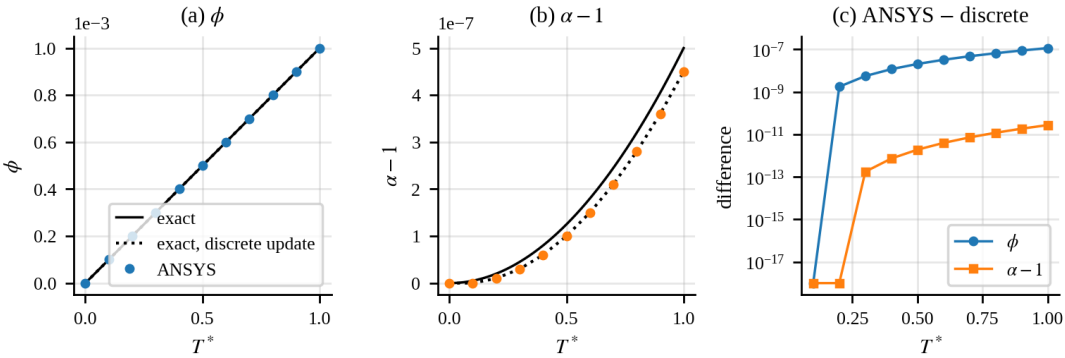
the element's field $\phi$ (NEM); $\beta = 0.02$ mm$^2/T^*$, converted (assumption)

**Added in this work**

$\dot{\alpha} = k_\alpha \phi$ at every Gauss point (Eq. 36) $\rightarrow \mathbf{F}_g \rightarrow$ stress

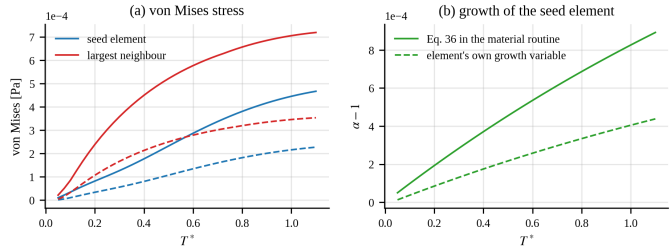
$\rightarrow$ The field formulation of the element is retained; only the growth law at the Gauss points is added.

# Part 2: verification against an exact solution




$\rightarrow$ Where $\phi$ has no gradient, $\phi = \sinh(k_\alpha t)$ and $\alpha - 1 = \cosh(k_\alpha t) - 1$: the coupled ANSYS solution agrees with the exact solution of the time integration to $3 \times 10^{-11}$ in $\alpha$.

# Part 2: growth-induced stresses with the published stiffness




Left: von Mises stress of the seeded element and its largest neighbour; right: $\alpha - 1$; solid: Eq. 36 at the Gauss points, dashed: the element's own growth variable.

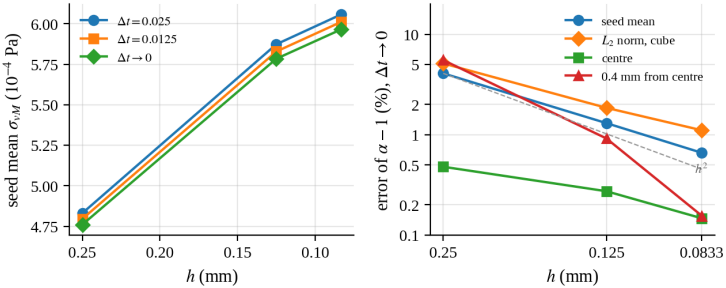
At $T^* = 1.1$ on $16^3$ elements:

| | |
|---|---|
| mean stress, seed | $-2.5 \times 10^{-4}$ Pa |
| mean stress, first layer | $+6.0 \times 10^{-5}$ Pa |
| mean stress, second layer | $+1.3 \times 10^{-6}$ Pa |
| von Mises, seed mean | $5.9 \times 10^{-4}$ Pa |
| $\alpha - 1$ in the seed | $8.9 \times 10^{-4}$ |

Stresses scale with $E = 10$ Pa.

$\rightarrow$ The seed is under compression and the surrounding material under tension, consistent with the ring of tension reported by Klempt et al. 2024.

# Part 2: convergence and stability




**Time step $\Delta t$ [$T^*$]**

first order; $\Delta t = 0.025$ gives an error of about 1 %

**Element size $h$ [mm]**

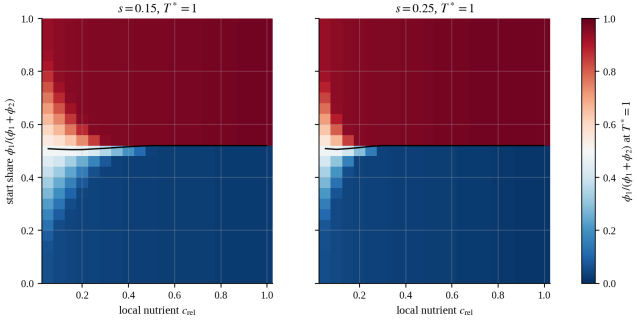
error of $\alpha - 1$ against a reference solution: 4.1 / 1.3 / 0.7 % on $8^3$ / $16^3$ / $24^3$; seed stress on $24^3$: 2 %

**Stability**

NEM limit derived (Bloch): 0.06 on $8^3$, 0.12–0.16 on $16^3$, $24^3$; confirmed by runs; Rudolf et al.

→ The error of the growth field is measured against a reference solution, the seed stress is 2 % from its converged value, and every run lies within the derived stability limit.

# Part 3: species composition at the Gauss points

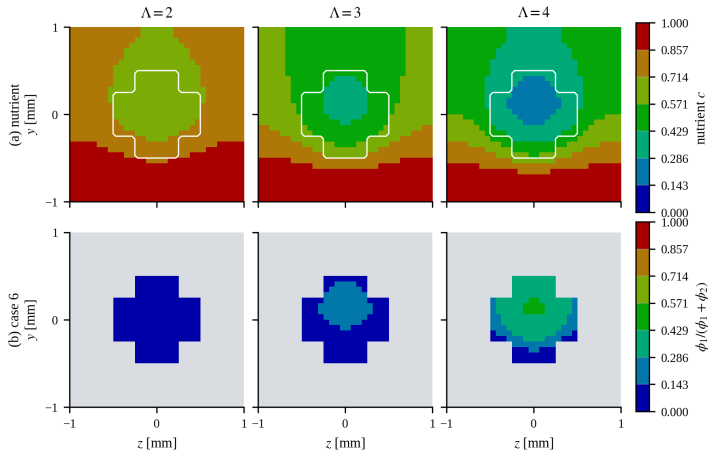



Case 6: final share of species 1 against its start share and the local nutrient level

- Two species, cases of Klempt et al. 2026; every call from ANSYS to the point model reproduced in Python bit for bit
- Share $\phi_1/(\phi_1+\phi_2)$ in the seed at $T^* = 1$: case 3 0.667, case 6 0.021
- Assumptions (not in the papers), with sensitivity: time link $s = 0.15$ (point model advances $s\,\Delta t$ per substep), cap $\phi_{\mathrm{cap}} = 0.9$, threshold $\phi_{\mathrm{min}} = 0.01$

$\rightarrow$ The coupled simulation retains the qualitative outcomes of the reference cases: coexistence (case 3) and competitive exclusion (case 6).

# Part 3: spatial heterogeneity under nutrient limitation




- Nutrient held at one face, consumed in the seed
- Strength: Thiele number $\Lambda^2 = gL^2/d$, with $L = 1$ mm the distance to the nutrient face
- Case 6, $\Lambda = 4$: share 0.03 near the nutrient, 0.44 inside
- Case 3 hardly changes (0.63–0.66)

$\rightarrow$ Nutrient limitation alone produces a spatially heterogeneous composition, which is absent for a constant $c^*$.

# Limitations

- **Verification, not validation**: no measured stress of a growing oral biofilm was available
- **No stress feedback**: the composition reaches the spreading only through the nutrient; the stress does not act back (as in Klempt et al. 2024)
- **Inactive front term**: no growth towards the nutrient in ANSYS; the front term of the element is inactive
- **Assumptions**: the time link $s$, $\phi_{\text{cap}}$, the conversion of $\beta$; each reported with its sensitivity
- **Calibrated parameters in the FE model**: one first run (CH) only; Part 3 uses two published cases
- **Symmetric A**: no directed effects; only part of **A** is determined; different strains in the commensal and dysbiotic data

$\rightarrow$ Each limitation is quantified in the thesis where possible; the results are verified, not yet validated.

# Conclusions and outlook

**Contributions**

**1 Calibration:** five species, four conditions; which interactions the data determine

**2 Growth law:** Klempt et al. 2024 in the IKM element, verified to $3 \times 10^{-11}$

**3 Composition:** two species at every Gauss point, qualitative outcomes of the reference cases retained

**Outlook (continuation at Keio University)**

- Five calibrated species in the finite element model
- Composition into stiffness, stress into growth (two-way)
- A working front term; explicit against implicit field update
- Posterior uncertainty carried to the stress

# References


- F. Klempt, M. Soleimani, P. Wriggers, P. Junker (2024), Biomech. Model. Mechanobiol. 23, 2091–2113: growth field, Eq. 34/36, Table 2
- F. Klempt, H. Geisler, M. Soleimani, P. Junker (2026), Arch. Appl. Mech. 96, 164: multi-species point model
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- P. Junker, J. Makowski, K. Hackl (2014), Contin. Mech. Thermodyn. 26, 259–268; K. Hackl (1997), J. Mech. Phys. Solids 45, 667–688
- T. Rudolf et al. (2025), Finite Elem. Anal. Des. 249, 104353: Neighbored Element Method
- N. Heine et al. (2025), Front. Oral Health: in-vitro data
- D. Feng et al. (2021), Bull. Math. Biol. 83, 48; M. Soleimani et al. (2023), Sci. Rep.: two-species models

# Appendix A: notation

**Growth field and mechanics (Klempt et al. 2024)**

| | |
|---|---|
| $\phi$, $\phi_{3D}$ | amount of biofilm (0 = void, 1 = dense) |
| $c$ | nutrient concentration |
| $\alpha$ | $\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$; growth is $\alpha - 1$ |
| $\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$ | elastic / growth split |
| $k_\alpha$ | growth rate ($10^{-3}\, T^{*-1}$) |
| $\beta$ | diffusion of $\phi$ (0.02 mm$^2/T^*$, converted) |
| $d$, $g$ | nutrient diffusivity, consumption |
| $r$, $k$ | front growth parameter, half-velocity constant |
| $E$, $\nu$ | 10 Pa, 0.49 (Klempt 2024, Table 2) |
| $T^*$ | normalised time |

**Point model (Klempt et al. 2026)**

| | |
|---|---|
| $\phi_i$, $\phi_0$ | volume fraction of species $i$, void |
| $\psi_i$, $\bar{\phi}_i$ | viability, living fraction $\phi_i\psi_i$ |
| $A_{ij}$ | symmetric interaction matrix |
| $c^*$, $\alpha^*$ | nutrient, antibiotic level |
| $\eta_i$ | dissipation parameters |
| $\gamma$ | multiplier of $\sum_l \phi_l = 1$ |

**Coupling (this work)**

| | |
|---|---|
| $\phi_1/(\phi_1 + \phi_2)$ | composition (share of species 1) |
| $s$ | point model advances $s\,\Delta t$ per substep |
| $\phi_{\text{cap}}$, $\phi_{\text{min}}$ | cap on the amount, start threshold |
| $\lambda = \beta\Delta t/h^2$ | stability measure |

# Appendix B: growth field and mechanics

**Kinematics.** $\mathbf{F} = \mathbf{F}_e \mathbf{F}_g$ with isotropic $\mathbf{F}_g = \alpha \mathbf{I}$. Stress comes only from $\mathbf{F}_e$. A body that can grow freely stays stress free; stress appears where growth is constrained.

**Material.** $E(\phi) = (\phi^2 + f)\, E$, $E = 10$ Pa, $\nu = 0.49$ (Klempt 2024, Table 2; their material is incompressible), $f = 10^{-3}$; decks in mm and MPa, stresses quoted in Pa.

**Field equations** (Klempt et al. 2024, Eqs. 34–36, without the bars):

$$\dot{\phi} = \underbrace{\beta \nabla^2 \phi}_{\text{diffusion}} + \underbrace{k_\alpha\, \alpha}_{\text{local source}} \underbrace{- \|\nabla \phi\| \frac{r\, c}{k + c} \mathbf{n}_{\nabla\phi} \cdot \mathbf{n}_{\nabla c}}_{\text{front, inactive in ANSYS}}, \qquad \dot{c} = d \nabla^2 c - g\, \phi\, c, \qquad \dot{\alpha} = k_\alpha\, \phi$$

- $\phi$ and $c$ are fields of the element, updated explicitly with the Neighbored Element Method; $\alpha$ is a state variable at the Gauss points
- Consumption $g\phi c$ as in the authors' implementation (printed: $g\phi$)

# Appendix C: point model and one coupled substep

**Point model** (Klempt et al. 2026), per species $i$, no space:

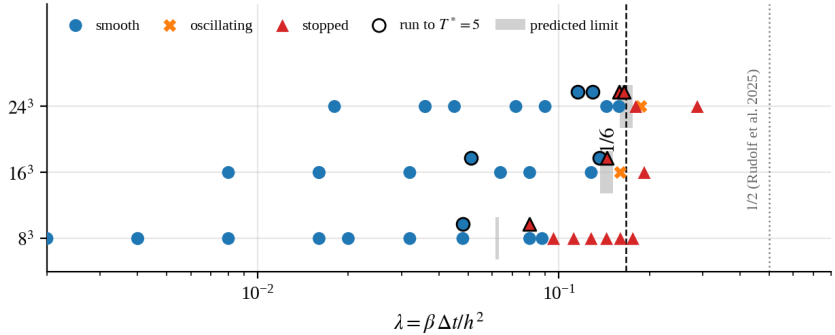
$$(\eta_{\phi,i} + \eta_i \psi_i^2)\,\dot{\phi}_i + \eta_i \phi_i \psi_i\,\dot{\psi}_i = c^* \psi_i \textstyle\sum_j A_{ij} \phi_j \psi_j - \gamma - (\text{barrier})$$
$$\eta_i\,(\phi_i \psi_i\,\dot{\phi}_i + \phi_i^2\,\dot{\psi}_i) = c^* \phi_i \textstyle\sum_j A_{ij} \phi_j \psi_j - \alpha^* b_i\,\psi_i - (\text{barrier})$$

with $\sum_i \phi_i + \phi_0 = 1$; the barrier keeps $\phi_i, \psi_i$ in $(0,1)$; implicit Euler and Newton, $\Delta t = 10^{-4}$.

**One substep of the coupled ANSYS run**

1. The element updates its field $\phi$
2. At each Gauss point: $\alpha_{n+1} = \alpha_n + k_\alpha \phi\,\Delta t$, $\mathbf{F}_g = \alpha \mathbf{I}$, stress and tangent
3. Two species, where $\phi_{3D} \geq \phi_{\min}$: scale the point model to $\min(\phi_{3D}, \phi_{\text{cap}})$, advance it by $s\,\Delta t$, store $\phi_i, \psi_i$ as state variables

# Appendix D: stability of the explicit field update




$\lambda = \beta\Delta t/h^2$; grey: the limit from a Bloch analysis of the NEM stencil on the lattice of the Gauss points (thesis, Appendix D). $h^2/(2\Delta t) \geq \beta$ of Rudolf et al. 2025 is the one-dimensional limit $(1/2)$; the seven-point stencil gives $1/6$; the NEM fit with its weight width $\beta^* = 0.2$ mm fixed in millimetres gives 0.06 on $8^3$ and, depending on which equidistant points enter the 30 neighbours, 0.12–0.14 on $16^3$ and 0.12–0.16 on $24^3$. Runs to $T^* = 5$ (black edge) confirm it; runs to $T^* = 1.1$ are too short to show an instability near the limit.

# Backup 1: stationarity conditions of the point model

$\dot{\Psi} + \Delta + C \to$ stat with respect to $\dot{\phi}_i$, $\dot{\psi}_i$, $\dot{\phi}_0$ and $\gamma$, where

$$\Psi = -\tfrac{1}{2} c^* \bar{\boldsymbol{\phi}}^{\mathrm{T}} \mathbf{A}\, \bar{\boldsymbol{\phi}} + \tfrac{1}{2} \alpha^* \boldsymbol{\psi}^{\mathrm{T}} \mathbf{B}\, \boldsymbol{\psi}, \qquad \Delta = \tfrac{1}{2} \dot{\bar{\boldsymbol{\phi}}}^{\mathrm{T}} \boldsymbol{\eta}\, \dot{\bar{\boldsymbol{\phi}}} + \tfrac{1}{2} \dot{\boldsymbol{\phi}}^{\mathrm{T}} \boldsymbol{\eta}\, \dot{\boldsymbol{\phi}}, \qquad C = \gamma\Big(\sum_{l=0}^{n} \phi_l - 1\Big)$$

gives, with $\bar{\phi}_i = \phi_i \psi_i$ and $\dot{\bar{\phi}}_i = \dot{\phi}_i \psi_i + \phi_i \dot{\psi}_i$,

$$\begin{aligned}
0 &= -c^* \psi_i (\mathbf{A} \bar{\boldsymbol{\phi}})_i + \eta_i \big( \dot{\phi}_i \psi_i^2 + \bar{\phi}_i \dot{\psi}_i + \dot{\phi}_i \big) + \gamma, \\
0 &= -c^* \phi_i (\mathbf{A} \bar{\boldsymbol{\phi}})_i + \alpha^* b_i \psi_i + \eta_i \big( \dot{\psi}_i \phi_i^2 + \bar{\phi}_i \dot{\phi}_i \big), \\
0 &= \gamma + \dot{\phi}_0, \qquad 0 = \textstyle\sum_{l=0}^{n} \phi_l - 1.
\end{aligned}$$

- $2n+2$ equations; implicit Euler and Newton, $\Delta t = 10^{-4}$; a logarithmic barrier keeps $\phi_i, \psi_i, \phi_0$ in $(0,1)$
- The second term of $\Delta$ is needed: the first alone gives a singular system
- Only the symmetric part of $\mathbf{A}$ enters $\Psi \Rightarrow \mathbf{A} = \mathbf{A}^{\mathrm{T}}$

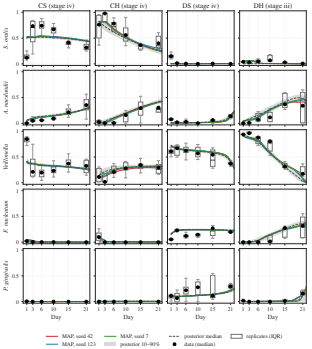
# Backup 2: symmetric interactions compared with an asymmetric model

| | gLV (L-BFGS-B) | Hamilton TMCMC |
|---|---|---|
| interaction matrix | asymmetric, 25 | symmetric, 15 |
| parameters | 30 | 20 |
| RMSE (four conditions) | 0.012–0.032 | 0.027–0.179 |
| uncertainty | point estimate | posterior, three seeds |
| $\sum_i \phi_i = 1$ | enforced afterwards | exact |

- The asymmetric model fits better, but has more parameters (30) than observations (25); its out-of-sample error was not assessed
- Directed effects (one species feeding another) cannot be represented; stated as a limitation

$\rightarrow$ Symmetry costs accuracy of the fit, but gives a posterior and exact conservation.

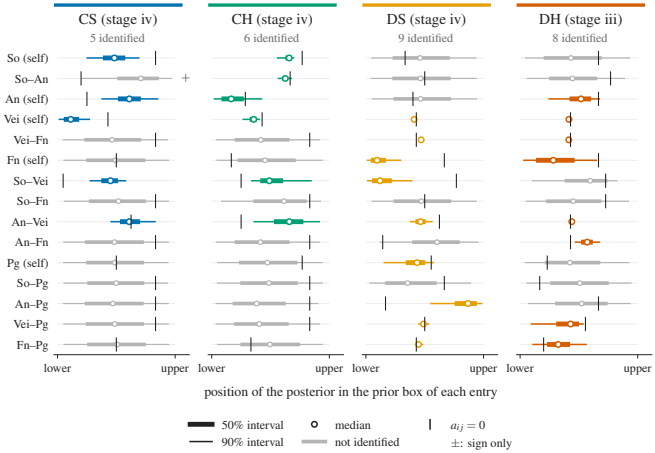
# Backup 3: four stages and three seeds




Stages: (i) composition; (ii) narrowed box; (iii) with the live fraction, full box; (iv) narrowed box. Accepted only if recomputed likelihoods agree, tempering reaches $\beta = 1$, every particle moves enough, and the seeds agree in $\max \ln \mathscr{L}$ and in the medians.

$\rightarrow$ Several runs that looked converged failed these checks and were repeated; DS and DH are from stage (iii).

# Backup 4: posterior distributions of the interactions




- Marginal posteriors of the 15 entries, drawn by their position in the prior box
- Identified: posterior SD below 0.8 of a uniform distribution on the box
- Sign only: 90 % interval excludes zero

# Backup 5: sensitivity to the two coupling assumptions

Share of species 1, $\phi_1/(\phi_1+\phi_2)$, at $T^*=1$

| | $s=0.05$ | $s=0.15$ | $s=0.5$ | $s=1$ |
|---|---|---|---|---|
| case 3, front | 0.638 | 0.672 | 0.701 | 0.744 |
| case 6, front | 0.230 | 0.014 | 0.014 | 0.014 |
| | $\phi_{\text{cap}}=0.85$ | 0.9 | 0.95 | |
| case 3, seed | 0.668 | 0.667 | 0.664 | |
| case 6, seed | 0.023 | 0.021 | 0.014 | |

- $s$: the point model advances $s\,\Delta t$ per substep; no paper links the time of Klempt et al. 2026 to $T^*$; $s=0.15$ maps its horizon onto $T^*=1$
- In case 6 the winner emerges for $s \gtrsim 0.15$ and does not change for larger $s$
- $\phi_{\text{cap}}$ hardly matters

$\rightarrow$ The outcome type is robust to both assumptions; the share in case 3 depends on $s$ by a few hundredths.

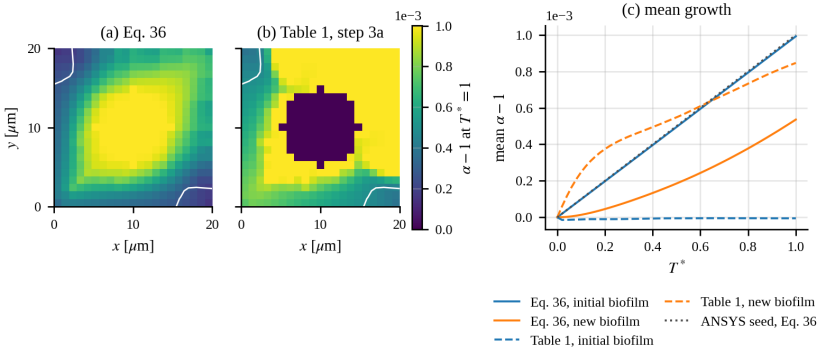
# Backup 6: sensitivity to the diffusion coefficient $\beta$

Mean stress in the seed at $T^* = 1.1$ [Pa]

| $\beta$ [mm$^2/T^*$] | 0.005 | 0.01 | 0.02 | 0.05 | 0.1 |
|---|---|---|---|---|---|
| $8^3$ | $-2.1 \times 10^{-4}$ | $-2.7 \times 10^{-4}$ | $-2.5 \times 10^{-4}$ | $-1.3 \times 10^{-4}$ | $-5.6 \times 10^{-5}$ |
| $16^3$ | $-3.9 \times 10^{-4}$ | $-3.4 \times 10^{-4}$ | $-2.5 \times 10^{-4}$ | $-1.2 \times 10^{-4\,\dagger}$ | $-3.8 \times 10^{-5\,\dagger}$ |

- $\beta = 0.02$: Table 2 of Klempt et al. 2024 ($\beta = 2$) converted to the 2 mm cube (assumption)
- $\phi$ diffusing out of the seed lowers its growth near the surface; the step in $\alpha$ between inside and surface loads the seed
- For $\beta \geq 0.02$ the diffusion length $\sqrt{\beta T^*} \geq 0.14$ mm is resolved by the mesh
- $^\dagger$ before the correction of the surface points

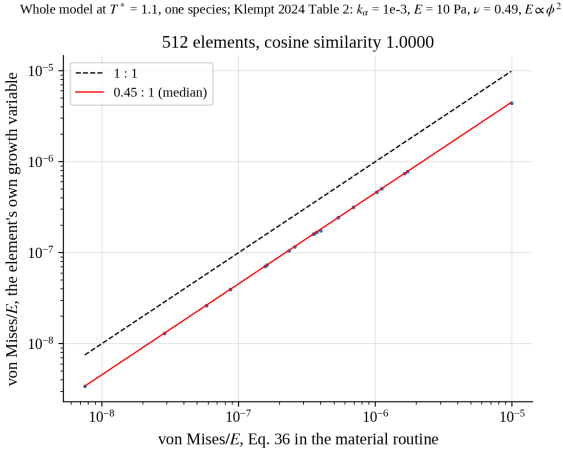
# Backup 7: the two growth laws of Klempt et al. 2024




The paper gives $\dot{\alpha} = k_\alpha \phi$ (Eq. 36) in the text and a different update in Table 1, step 3a. On the reproduced field of test case 4.1 they differ: a seed full from the start grows only under Eq. 36.

→ I follow Eq. 36, as the text of the paper and the element do.

# Backup 8: Eq. 36 against the element's own growth variable




- Whole model, 512 elements: the two von Mises fields have the same pattern, cosine similarity 1.0000
- The element's variable gives 0.455 times the stress of Eq. 36, which is $\frac{1}{2} \times \frac{10}{11}$: the mean over both species fields, one of them unused, and a lag of one substep
- With Eq. 36 at the Gauss points, the growth is counted once per increment

$\rightarrow$ Same spatial pattern; the difference in magnitude is fully explained by how the element's variable is formed, not by the mechanics.

# Backup 9: mesh and incompressibility

- Python, same mechanical problem on $8^3$, $16^3$, $32^3$ linear hexahedra: seed averages on $8^3$ are 1.6–1.75 times those on $16^3$ and $32^3$, which agree with each other; corner peaks do not converge (singular corners) $\Rightarrow$ seed averages are quoted, not peaks
- Mean-dilatation (B-bar) integration, as SOLID185 with KEYOPT(2)= 0: mean stress $-3.8$ and $-4.1 \times 10^{-5}$ Pa for $\nu = 0.49$ and 0.499 on $8^3$ $\Rightarrow$ no volumetric locking
- Full $2 \times 2 \times 2$ integration would overestimate the mean stress by a factor of 3.7 on $8^3$ and 2.3 on $16^3$ at $\nu = 0.49$
- ANSYS: $16^3$ and $24^3$ with every boundary condition mapped by position; $32^3$ exceeds the memory of the workstation

$\rightarrow$ The quoted seed averages are mesh-converged on $16^3$ and $24^3$; B-bar avoids locking at $\nu = 0.49$.

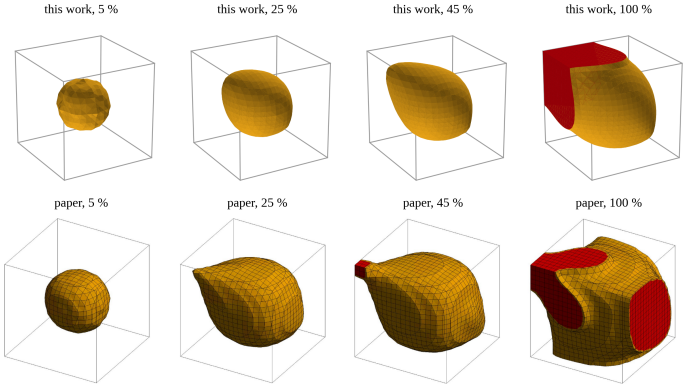
# Backup 10: correction of the surface points of the element

- In the element, $\phi$ at the surface points was not updated and stayed 0: every face acted as $\phi = 0$, where the model has zero flux
- Corrected: each surface point takes the distance-weighted mean of its neighbouring Gauss points; all thesis runs repeated

| runs | seed von Mises | seed pressure | $\alpha - 1$ |
|---|---|---|---|
| $\beta = 0.02$ ($8^3$, $16^3$, one and two species) | $\leq 0.01\,\%$ | $\leq 0.13\,\%$ | unchanged |
| $\beta = 0.05$, $8^3$ | $+0.3\,\%$ | $-3.5\,\%$ | $0.02\,\%$ |
| $\beta = 0.1$, $8^3$ | $+6.7\,\%$ | $-30\,\%$ | $0.5\,\%$ |

$\rightarrow$ The main results change by less than $0.5\,\%$; only runs in which $\phi$ reaches the faces change noticeably, and their corrected values are used.

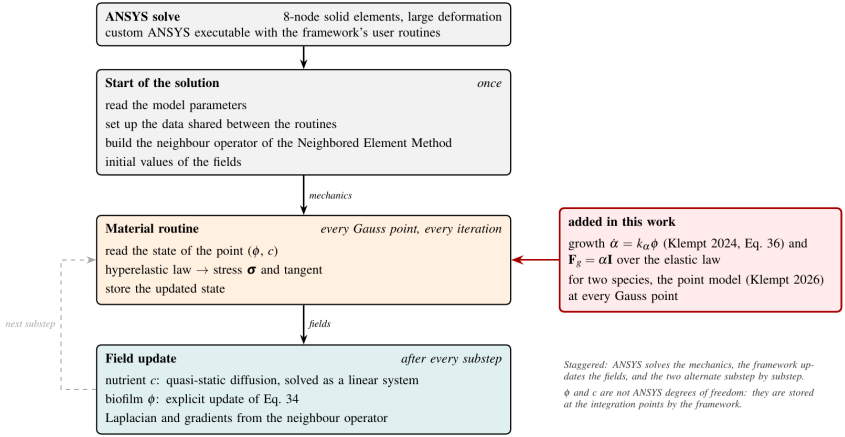
# Backup 11: reproduction of Klempt et al. 2024, test case 4.1




Python reproduction with the parameters of Table 2; consumption $g\phi c$ as in the authors' implementation (printed: $g\phi$), front term with an isotropic share.

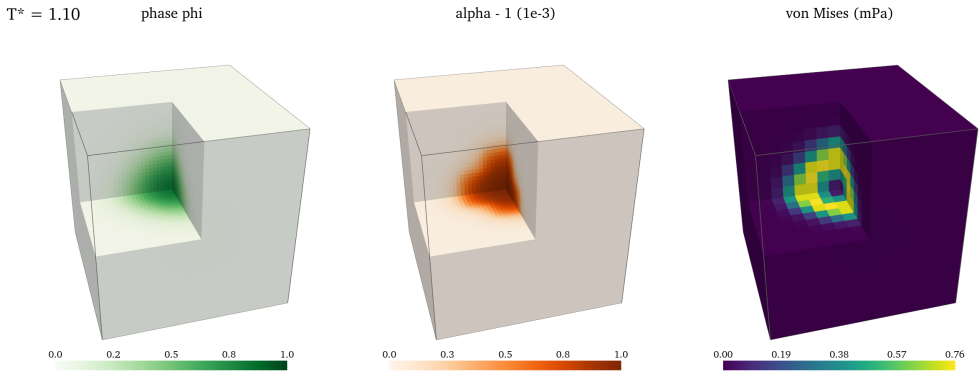
→ Case 4.1 is reproduced with stated departures from the printed equations (thesis, Appendix C).

# Backup 12: solution loop of the element framework




→ The material routine is called at every Gauss point and iteration; the fields $\phi$ and $c$ are advanced after each substep.

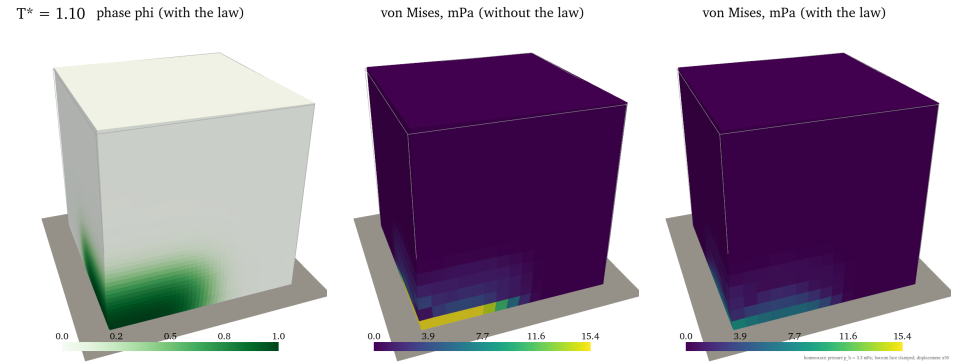
# Backup 13: the verification problem in 3D




Seed in the centre of the 2 mm cube, one octant cut away; $\beta = 0.02\,\mathrm{mm}^2/T^*$, $k_\alpha = 10^{-3}/T^*$, $E = 10\,\mathrm{Pa}$. Field: reference solution on $48^3$; stress: $16^3$. Video: `assets/anim_wp3_seed_3d.mp4`.

$\rightarrow$ Seed von Mises $6.19 \times 10^{-4}$ Pa at $T^* = 1.1$, within 0.2 % of the value in the thesis; the stress forms a ring at the edge of the seed.

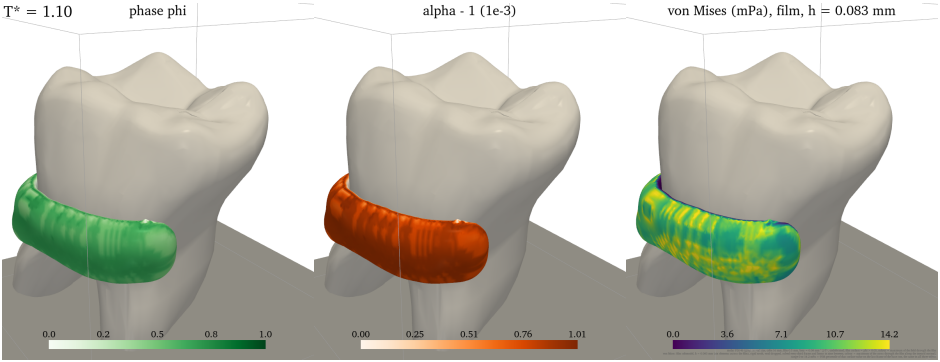
# Backup 14: outlook, stress at a wall and a homeostatic-pressure law




Not part of the thesis: the same model with the seed in a corner on a clamped face, without and with growth scaled by $g = \min(1, \max(0, 1 - p/p_h))$; $p_h = 3.3$ mPa chosen by me. Video: `assets/anim_law_3d.mp4`.

→ In the biofilm next to a fixed surface the stress is 10 to 12 times the seed stress of the centre case (band mean, converged to 1 %); limiting growth by pressure removes the concentration. Continuation at Keio.

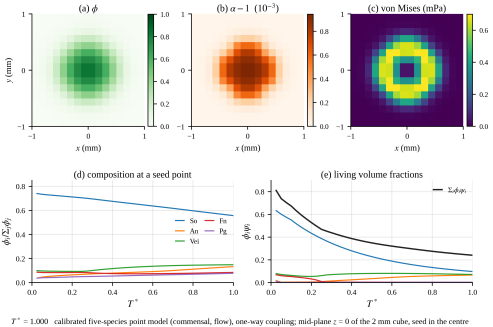
# Backup 15: outlook, a biofilm on a real molar




Not part of the thesis: the same growth model on a lower first molar at real size (public surface model, CC BY 4.0), a 0.5 mm film near the gum line. Resolved with six elements across the film, the film mean von Mises stress is 3.3 mPa ($\pm 1\,\%$), highest in the 0.1 mm next to the tooth.

$\rightarrow$ Thin films on real teeth need a film-resolving mesh; with it the stresses converge. Continuation at Keio.

# Backup 16: the calibrated five-species model in the element, over time




$T^* = 1.000$ calibrated five-species point model (commensal, flow), one-way coupling; mid-plane $z = 0$ of the 2 mm cube, seed in the centre

The run of Section 4 (commensal flow condition, final MAP, Day-1 start, one-way coupling) on the finer $16^3$ mesh, 40 output times up to $T^* = 1$. At $T^* = 1$: seed mean $\alpha - 1 = 6.56 \times 10^{-4}$, von Mises $5.74 \times 10^{-4}$ Pa ($8^3$ in the thesis: $6.75 \times 10^{-4}$, $4.68 \times 10^{-4}$ Pa); shares within 0.01 of the $8^3$ run. Video: `assets/anim_cal5_ch.mp4`.

→ *S. oralis* falls from 0.74 to 0.56 and the living fraction from 0.81 to 0.24, while growth and stress do not depend on the composition (one-way coupling).